A small package review

Example: fictional bench fixture BF-100, revision B. All source content was created for this demonstration with AI assistance. It is not client work or a manufacturing package.

S1 - Fictional customer baseline

Document DEMO-S1, revision B, dated 2026-10-08. For this exercise only, treat this baseline as the controlling source. No real customer approval is represented.

Part / description	Revision	Qty per fixture	Material / definition
BF-101 / Base plate	B	1	Aluminum 6061-T6
BF-102 / Spacer	A	2	Aluminum 6061-T6
HW-103 / Socket screw	A	4	Purchased M6 x 16; hardware drawing excluded

Required package: one drawing PDF and one STEP file for each fabricated part (BF-101 and BF-102), plus a BOM. Hardware has no drawing or STEP requirement in this example. Geometry, fits and fastener suitability are outside this exercise.

S2 - Candidate BOM excerpt

Document DEMO-S2, fixture BF-100, revision B. Quantities below are as supplied in the fictional candidate.

Row	Part	Revision	Qty per fixture
1	BF-101	B	1
2	BF-102	A	1
3	HW-103	A	4

Review question: does the candidate package agree with S1, and what evidence is still missing? Preserve the supplied candidate and record discrepancies without silently correcting it.

Candidate evidence

S3 - Drawing title-block excerpt

The following is fictional transcribed drawing metadata, not a CAD-generated drawing. This sample demonstrates document comparison only.

Field	Observed value
Listed filename	BF-101_B.pdf
Part number in title block	BF-101
Revision in title block	A
Material note	Aluminum 5052-H32
Sheet	1 of 1

S4 - Complete fictional candidate inventory

For this exercise, the inventory below is declared complete. These names are source data in this PDF; no matching CAD files or ZIP archive are supplied.

Entry	Listed file	Role
1	BF-100_B_BOM.csv	Candidate BOM; content shown in S2
2	BF-101_B.pdf	Drawing metadata shown in S3
3	BF-101_B.step	Listed neutral model; contents unavailable

Evidence limits

The reviewer can compare S1-S4. The reviewer cannot inspect native geometry, STEP contents, dimensions, tolerances, material properties inside CAD, archive bytes or real approval records. A filename alone does not establish its contents or revision.

Customer-facing result: a short findings register that identifies each conflict, cites the source and states the decision or evidence needed to close it.

Five discrepancies to resolve

Disposition: candidate does not match the fictional baseline. Hold this exercise package for correction and re-review. This is a document-consistency finding, not engineering or manufacturing approval.

ID / finding	Source evidence	Proposed resolution / owner
D01 - Drawing revision conflict	S1: BF-101 rev B. S3 title block: A, despite filename B.	Design owner supplies the intended revision and matching drawing. Verify contents, not just the renamed file.
D02 - Material conflict	S1: BF-101 6061-T6. S3: 5052-H32.	Design owner confirms material and supplies a consistent drawing or approved baseline change.
D03 - Quantity conflict	S1: two BF-102 spacers per fixture. S2 row 2: one.	BOM owner reconciles quantity against the approved assembly before updating the BOM.
D04 - Required drawing absent	S1 requires BF-102 drawing. S4 has no BF-102 drawing entry.	Package owner supplies the drawing; reviewer checks part, revision and content.
D05 - Required STEP absent	S1 requires BF-102 STEP. S4 has no BF-102 STEP entry.	Package owner supplies the STEP; geometry and identity require an independent CAD check.

What matched within the stated scope

All three BOM part numbers and revisions agree with S1. BF-101 and HW-103 quantities agree. Listed BF-101 filenames carry revision B, but D01 shows why filenames are insufficient. No hardware drawing is flagged because S1 excludes it.

Unverified - not a pass

Native CAD, dimensions, tolerances, fits, actual STEP geometry, real file presence/ZIP integrity and release authority were not available. These require additional inputs and a separately agreed check. No source was modified and none of the five findings is closed.

Delivery method: AI-assisted comparison with explicit source references and a checked report. This original demonstration is evidence of the reporting format, not a claim of prior paid work or native CAD execution.